Week 2 Update

GM2 - Aqua Solutions

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Recap – Week 1

- **Prototype and Trial Challenges:** Sensor degradation, fuzzy logic algorithm collapsed, technical complexity.
- **Approach: TinyML** algorithm trained on public data → Edge AI device.
- **Objective:** Detection & advisory (instead of purification).
- **Mechanical Improvements:** Can we reduce perpetual exposure to water to mitigate the effects of sensor damage?

Initial steps

- **Literature review —**
  - mapped the research space
  - identified gaps in sensor fusion and community-sourced labelling

- **Algorithm selection —**
  - reviewed ML methods applied to similar datasets under similar constraints
  - selected XGBoost

- **Public data trial —**
  - ran XGBoost on Kaggle water quality datasets
  - poor performance due to synthetic data, class imbalance (95% HIGH risk), and missing bacterial concentration values

Literature Review

Explored	Unexplored
Single-parameter sensors	Multi-sensor fusion for pathogen inference
Single-species detection (E. coli in isolation)	ML based risk inference from proxies
Lab-based validity only	Field deployment robustness & drift correction
Low-cost IoT hardware (ESP32, Arduino)	Illness reports as training labels

XG Boost

🌱 One decision tree = a few yes/no questions about the water

🌲🌲🌲 XGBoost builds hundreds of trees in sequence

🎯 Each tree corrects the mistakes of the last

✅ Together they detect patterns no single rule could catch

⚡ **Fast & lightweight** — less computationally expensive + smaller dataset requirements than RNNs

🔧 **Sensor failure tolerant** — naturally handles missing data, sensor drift and saturation

📊 **Probabilistic labelling** — more illness reports = higher confidence, avoids false alarms

📡 **Edge-ready** — compressed model runs on-device, no internet required

Challenges - Data

Synthetic Data

Do not reflect real-world variability.

Incomplete Datasets

Missing values. Lacking bacterial concentration.

Inappropriate Datasets

Different contamination mechanisms. Different climates.

Bacterial Concentration ≠ Disease

Some E. Coli strains are good, others bad.

Skewed Results

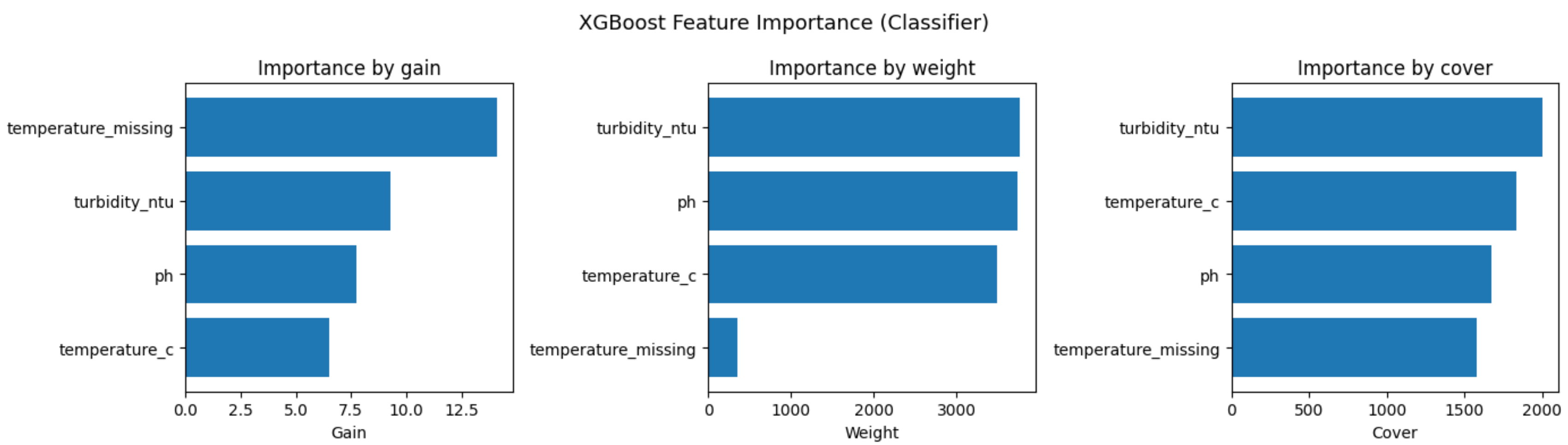
Model can achieve a small loss by always predicting HIGH risk (95%).

No Paired Outbreak Data

When does contamination mean disease?

Results and Learning

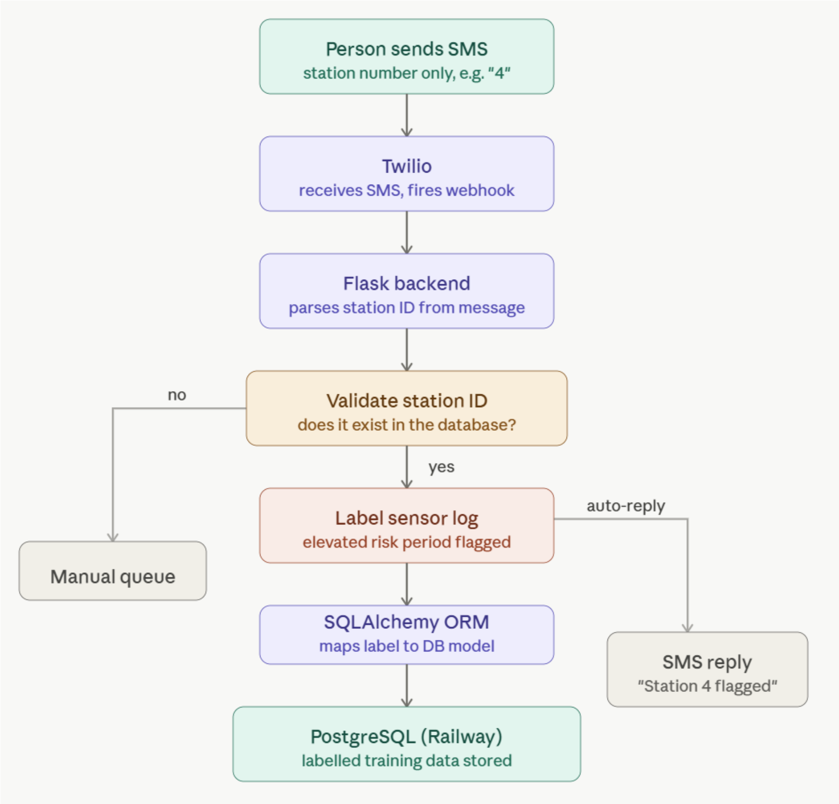
- **XGBoost Regression**: Literature Approved
  - R^2 = - 0.17 (performs worse than average)
  - Failed to beat 95% threshold.
  - Adding time of year helped improve prediction (wet vs. dry season).



New Objective

Develop an open-source, modular dataset collection service enabling us (or others) to leverage modern ML techniques for water quality issues.

System Architecture



- **Sensor unit** – multi-parameter probes on boreholes logging at regular intervals
- **Community flagging** – illness reports via SMS or website, one text labels data
- **Remote ML** – XGBoost trained on labelled sensor data combined with local weather recordings

Sensor Unit

Silva et al. (2022) — *Advances in Technological Research for Online and In Situ Water Quality Monitoring: A Review.* Sustainability, 14, 5059.
https://doi.org/10.3390/su14095059

Choice of Sensors…

- **pH** — bacterial activity shifts acidity before contamination is visible
- **Turbidity** — faecal matter clouds water; one of the earliest warning signals
- **Chlorine residual** — drops when disinfection is failing or overwhelmed
- **ORP** — falls sharply when organic contamination enters the system
- **UV absorbance (254nm)** — faecal organic matter absorbs UV; high absorption = high risk

Functionality…

- **Takes readings at regular intervals**— balances data granularity against power and connectivity costs
- **Transmits over GSM/2G** — wide rural coverage in Zimbabwe; LoRaWAN an alternative for clustered boreholes within 15km of a gateway
- **Readings stored locally, transmitted twice daily** — reduces data costs and tolerates intermittent connectivity

HealthCare in Zimbabwe

- **October 9th, 2025:** Establishment of the Public Health Institute of Zimbabwe (PHIZ)
- Coordination centre → data-driven approaches.
- **Goals:** Strengthen emergency preparedness, multi-sector operations.
- **Village Health Worker Program:** 21,000 VHWs as of 2026.
- **Mission:** Education, sanitation, vaccines, awareness.
- **Impilo**: Web-based electronic healthcare system (HIV, TB, maternal health)

Reporting Unit

Peletz et al. (2018) — *Why do water quality monitoring programs succeed or fail? A qualitative comparative analysis of regulated testing systems in sub-Saharan Africa.* International Journal of Hygiene and Environmental Health, 221, 907–920. https://doi.org/10.1016/j.ijheh.2018.05.010

Structure...

- **Website** — structured reporting and data analysis, used when connected (VHWs at a clinic, administrators reviewing borehole data)
- **SMS** – field reporting, works anywhere with 2G signal (VHWs in the field, community members)

Design Principles...

- **Minimal friction** — one text with just the station number, e.g. ”station 4", unambiguous, no NLP needed
- **Universal Access** — SMS over 2G, no internet or app required, anyone with any phone can report
- **Closed loop** — automated reply confirms the report was received, keeps the community engaged and reporting
- **Generating safe labels** – results will improve dramatically if safe labels come from confirmation and not the absence of unsafe labels

Edge AI

Structure...

- **Trained remotely** — XGBoost trained on cloud using labelled sensor + community data
- **Compressed & Deployed** – Model pushed to microcontroller, no hardware change required
- **On-device inference –** outputs risk score + confidence with every reading, no internet required

Training pipeline...

- **Cold start** — Manual surveys by VHWs for a preliminary dataset combined with threshold rules (e.g. chlorine < 0.2 mg/L)
- **Improves over time** – model retrains as labelled data accumulates

Conversation with Allen Chafa

Q: How do we know the product will perform in **various environments**? Will we have to retrain it at every location?

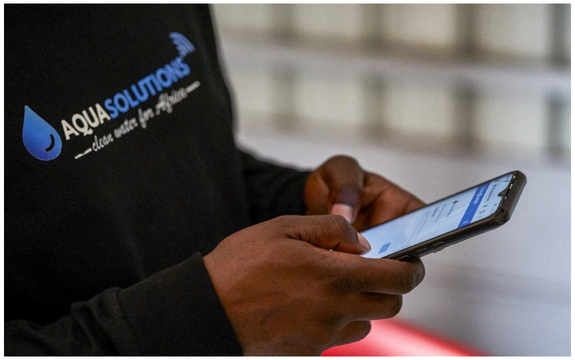
A: Base Model → Deploy → **Tailored Data** → Retrain/Refine

Q: Will we consider **tap water** or only boreholes?

A: Both! Readings at the household-level are more direct and can deal with **point-of-use recontamination**. However, **80% of water consumption** in Harare is from **boreholes** (over 10,000 in the city).

Q: 2G or LoRa? **2G**.

Feedback: Individuals have **limited incentive or understanding** of the importance for reporting consistently. Village Health Workers (VHWs) are the more reliable data source for illness labelling. Would requiring simple **training program**.



Further Challenges

Decoupling
How do we know whether an illness report is water-borne or propagated otherwise?

Access
VHWs provide lots of service to some areas, while others are scarcely reached.

Training
Education and training must highlight the importance of data collection, despite its indirect impact.

Incentive Structure
How can we convince people to use our SMS messaging system?

Microcontroller
Parts must be sourced locally for sustainability, repairability, and compatibility.

Energy and Infrastructure
Power shortages disrupt communications.

Next Steps - Software

- **Set up XGBoost training pipeline on the PostgreSQL schema**
  - Train on synthetic baseline data; retrain as field data arrives
- **Integrate model output into app UI**
  - Risk score + confidence displayed per station per reading
- **Introduce safe labelling mechanism**
  - Weekly "All well? Reply Y" SMS to registered users
- **Design the complete communications pipeline**
  - From sensor data to model download, need to define the flow of information.

Next Steps – Community Engagement

- **Considerations:**
  - **Compliance** (HL7 FHIR): Standard for **digital healthcare**.
  - Review of current health recording/platforms → **Integration?**

- **With Allen Chafa:**
  - Speak to **industry partners** about potential **applications**.
  - Speak to **healthcare workers** about a feasible, engaging **UX**.

- Digital Form
- Photo of Existing Documents
- Vocal Report
- SMS

Questions?

Thank you!